

3	(a)		The electric field strength at a point is the <u>electric force per unit positive charge acting on a</u> small stationary charge placed at that point. [B1]
	(b)		since particle is undeflected, net force on particle is zero OR gravitational force is balanced by the electric force [B1] $mg = q \frac{V}{d}$ $m = (3.2 \times 10^{-19}) \frac{100}{(0.06) \times 9.81} \text{ [C1]}$ $m = 5.4 \times 10^{-17} \text{ kg [A1]}$
	(c)	(i)	As there is now a <u>constant</u> vertical/ upward net force acting on the particle [M1] and the particle has a <u>constant</u> horizontal velocity [M1], the particle will now travel in a <u>parabolic path</u> [A1]
		(ii)	$F = qE$ $= (3.2 \times 10^{-19}) (200/0.060) \text{ [C1]}$ $= 1.1 \times 10^{-15} \text{ N [A1]}$
		(iii)	$F_{\text{net}} = (1.1 \times 10^{-15}) - (5.4 \times 10^{-17})(9.81) = 5.7 \times 10^{-16} \text{ N [C1]}$ $a = \frac{5.7 \times 10^{-16}}{5.4 \times 10^{-17}} = 10.6 \text{ m s}^{-2} \text{ [A1]}$ upwards [A1]

4	(a)		The maximum displacement of the mass from the equilibrium position/fixed
---	-----	--	--